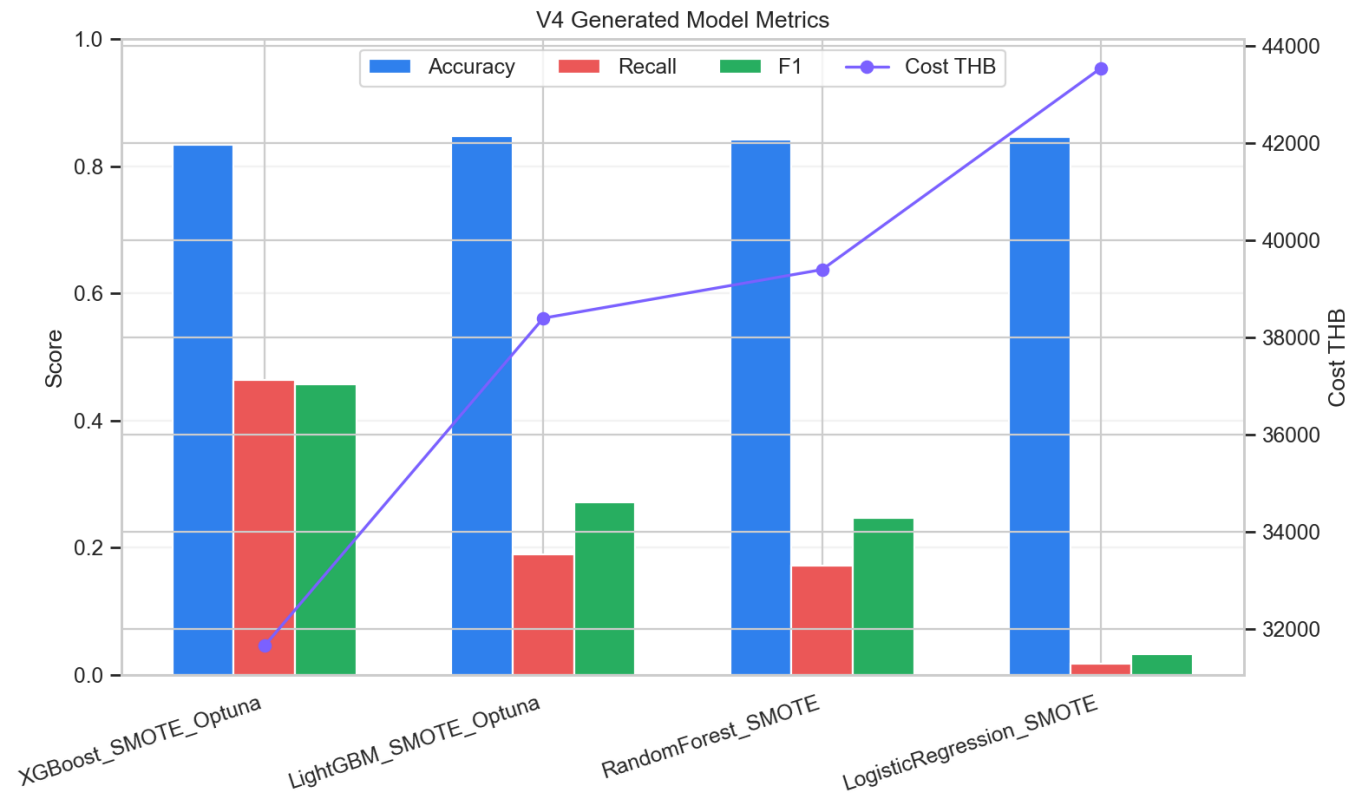


V4 Generated End-to-End Report

Generated imbalance data, cleansing, EDA, feature engineering, SMOTE, model training, Optuna tuning, and SHAP analysis.

Source rows	5000
Synthetic rows added	4700
Final rows	9700
Return rate	0.1500
Best model	XGBoost_SMOTE_Optuna
Accuracy	0.8345
Recall	0.4639
F1	0.4569
AUC	0.8538
Cost THB	31,650



model	threshold	accuracy	precision	recall	f1	auc	cost_thb
XGBoost_SMOTE_Optuna	0.3400	0.8345	0.4500	0.4639	0.4569	0.8538	31,650
LightGBM_SMOTE_Optuna	0.5800	0.8474	0.4783	0.1890	0.2709	0.8526	38,400
RandomForest_SMOTE	0.5000	0.8423	0.4348	0.1718	0.2463	0.7661	39,400
LogisticRegression_SMOTE	0.6400	0.8459	0.2778	0.0172	0.0324	0.7159	43,550